5. Running thread · Reading code What do $g(3)$ and $g(-5)$?

```
def g(x):
    if x > 0:
        return 2*x
    else:
        return -x
```

Sheet 38 Mon. 1st Feb. 2027 Sign of a product or quotient; sign tables

1. Warm-up · Flash Sign of $2 \quad x-4$?

2. Core · Method Draw the sign table of $(x-1)(x+3)$.

3. Core + · Method Solve $\dfrac{x+2}{x-1} \geqslant 0$.

4. Back · Flash 32-card deck: probability of drawing a heart? a king?

5. Running thread · Refute Disprove: "for every real number x, $x^2 \geqslant x$ ».

Proportions and changes

Sheet 39 Tue. 2 Feb. 2027 Proportions and percentages; proportion of a proportion

1. Warm-up · Flash Calculate 15% of 240.

2. Core · Method 30% of pupils are boarders and 40% of boarders dance. Which proportion of the students are boarders who dance?

3. Core + · Problem A lycée has 800 students, 55 % of whom are girls; 20 % of the girls take Latin. How many girls take Latin?

4. Back · Flash 3 notebooks cost 4.50 €. Price of 7 notebooks?

5. Running thread · True/false True or false? Justify. The number -2 is a solution of the equation $3 \quad x+1=-5$.

Sheet 40 Thu. 4 Feb. 2027 Absolute change, multiplier, rate of change

1. Warm-up · Flash A price goes from 80 Ar to 92 Ar. What is the absolute change?

2. Core · Method For the same change, give the rate of change and the multiplier.

3. Core + · MCQ A 20 % decrease corresponds to the multiplier:

A. 0,2 **B.** $-0{,}8$ **C.** 1,2 **D.** 0,8

4. Back · Flash Temperatures -2, 0, 3, 5, 9 (in °C). Mean and range?

5. Running thread · Flash Calculate the volume and total surface area of a cube with edge 3 cm.

Sheet 41 Mon. 8 Feb. 2027 Successive percentage changes

1. Warm-up · Flash Calculate $1.2 \times 0{,}9$.

2. Core · Method A 20% increase is followed by a 10% decrease. What is the overall rate?

3. Core + · True/false True or false? Justify. A 50 % rise followed by a 50 % fall brings back to the initial value.

4. Back · Flash Calculate $x^2 - x$ for $x = \dfrac{1}{2}$.

5. Running thread · Write up $ABCD$ is a parallelogram such that $AC = BD$. What type is it? Justify by one sentence.